

CLES 101 – September 2nd, 2026

Atmospheric Composition

Ch 1

No Office Hours Tomorrow

Today: Atmospheric basics

- “History” of Earth’s atmosphere
- Trace gases
- Mixing ratio vs number density vs mass density
- 0th law of thermodynamics

Commutativity of equilibrium.

Associated reading: Chapter 1- Characteristics of the atmosphere

Your to-do list *from last class*

~~For our next class:~~ For today:

1. Double check you can access the Canvas site *and* the CLES101 Shared Google Drive
2. Let me know if you do not have regular access to a device where you can work on Google Colab-hosted assignments
3. Get access to our textbook (Marshall & Plumb)

For next week:

1. Read the Southern California Listening Report (On Drive, under 'non_assignment_readings' → Week1_Southern_California_Listening_Sessions.pdf)
2. Complete the 'Getting to Know You Survey' (On Canvas, under 'Quizzes' → 'Surveys' (Week 1: Getting to know you and some reflections) *(ideally, for Monday)*)
3. Read Marshall & Plumb Chapter 1
4. Try your first Foundational Quiz *(ideally, before Wednesday)*



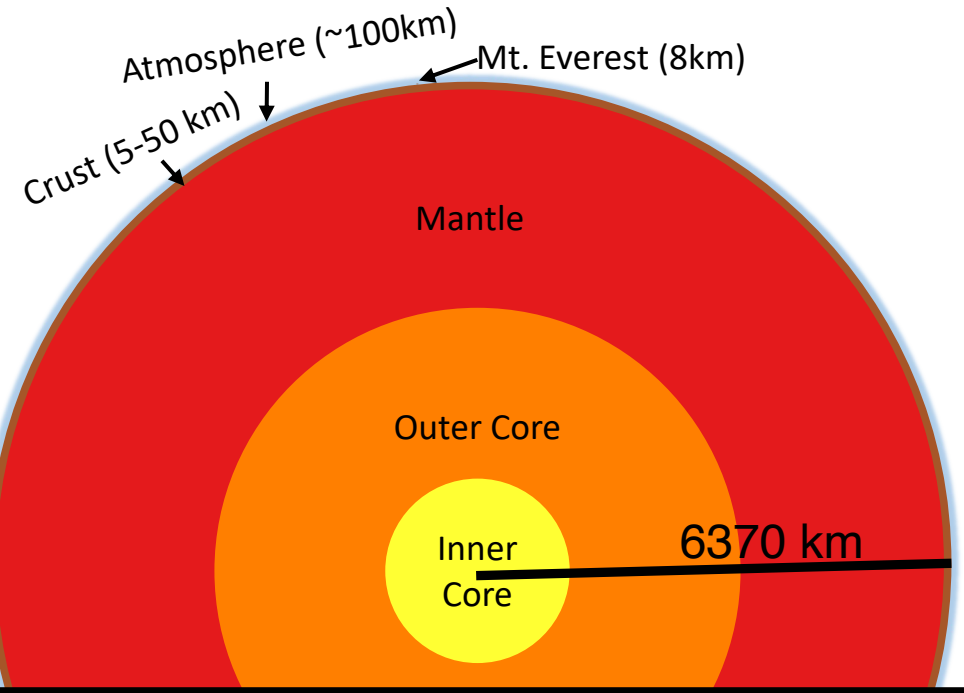
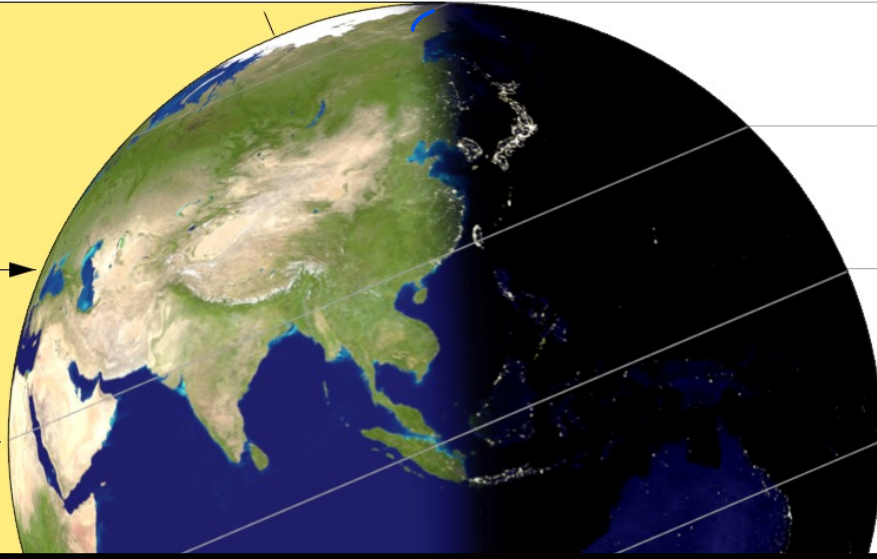
For your notes: What are the key properties of our planet that are being highlighted here? Why do they matter?

Refresher on... Earth!

Earth Axis 23.5°

Credit: NASA's Picture Dictionary

Sun rays



A Big idea:

B Big idea:

Evolution of Earth's Atmosphere

Atmosphere #1

~4.5 billion years ago



Key concepts:

- Sometimes called the proto earth stage

-
- We think the early atmosphere was mostly made of helium and hydrogen
 - Why do we think that?

-
- What happened to “Atmosphere #1”?
-
-



faint young sun - less sun but a lot of CO_2 , but
carbon cycle

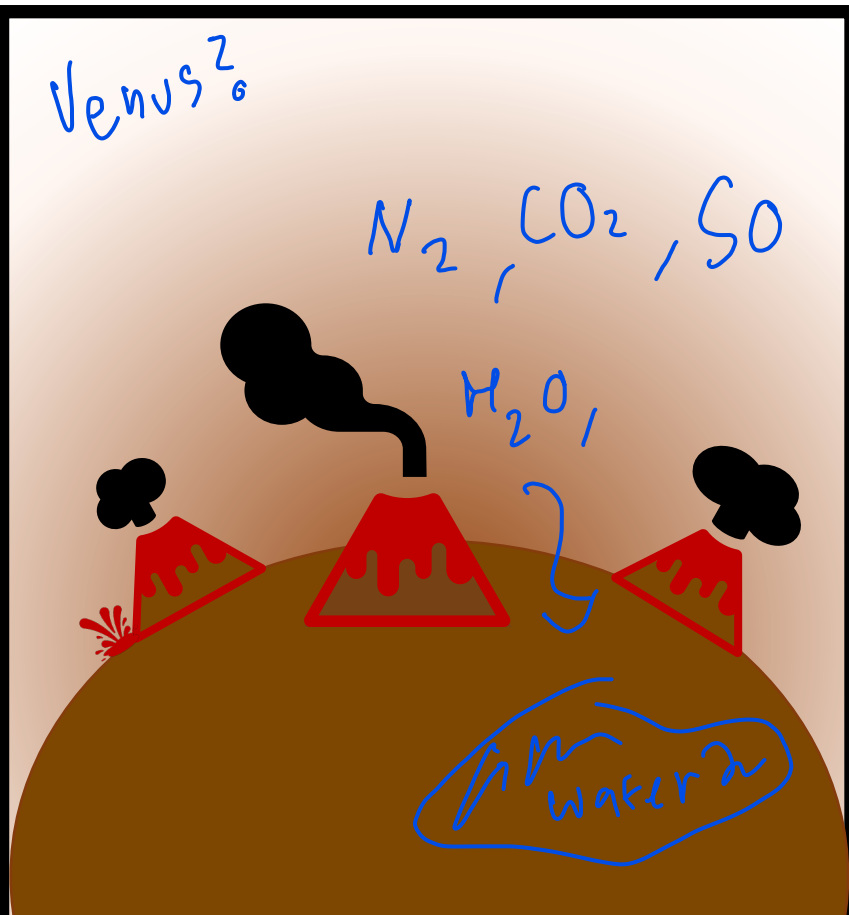
Carbonate water $\rightarrow$ Carbonate
rocks to draw water

Evolution of Earth's Atmosphere

Volcanism from mantle convection

Atmosphere #2

4.5-2.5 billion years ago



Key concepts:

- What was the planet like ~4.5 billion years ago?

-
- Where did this atmosphere come from?

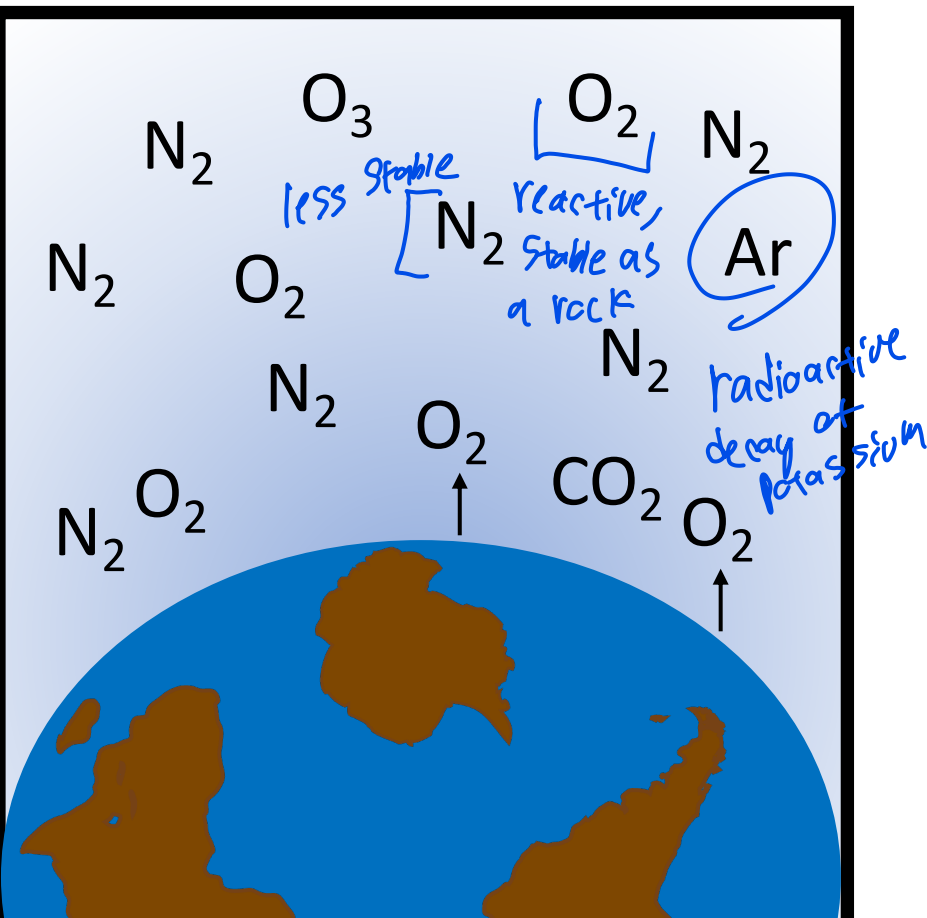
-
- What were some of the major components of this atmosphere? _____, _____, and _____.
 - Why do we think that?

-
- Why isn't Earth like Venus and how come Earth started to cool down? _____
-
-

Evolution of Earth's Atmosphere

Atmosphere #3

~2.5 billion years ago - present



Key concepts:

- What special thing happened 2.5 billion years ago?

phytoplankton blooms

- What did that add to our atmosphere?

O_2

- What did that remove from our atmosphere?

CO_2

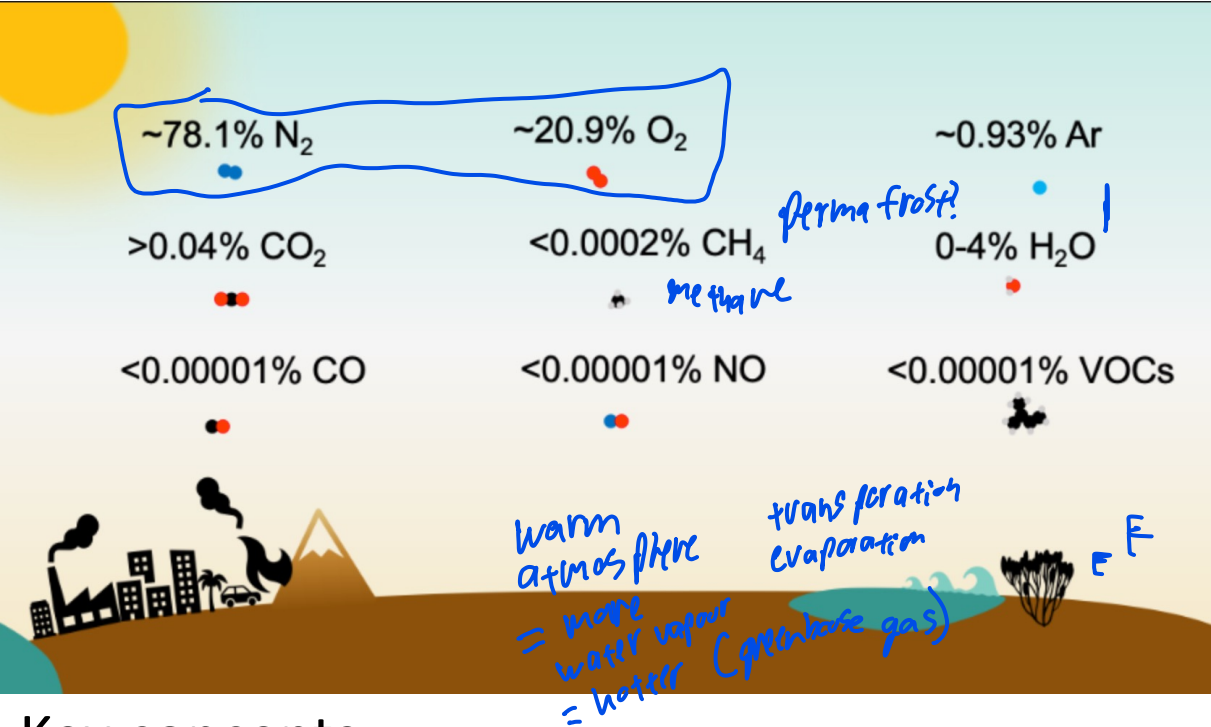
- What happened when O_2 built up?

Ozone → less heat/radiation

- What did that allow to happen on land

life

Composition of our present atmosphere



Key concepts:

- Where did the Ar come from?

Na decay

- Where does CO_2 come from?

- Where does CH_4 come from?

- Where does H_2O come from?

- Where does NO come from?

Key concepts:

- Why so much N_2 over of O_2 ?

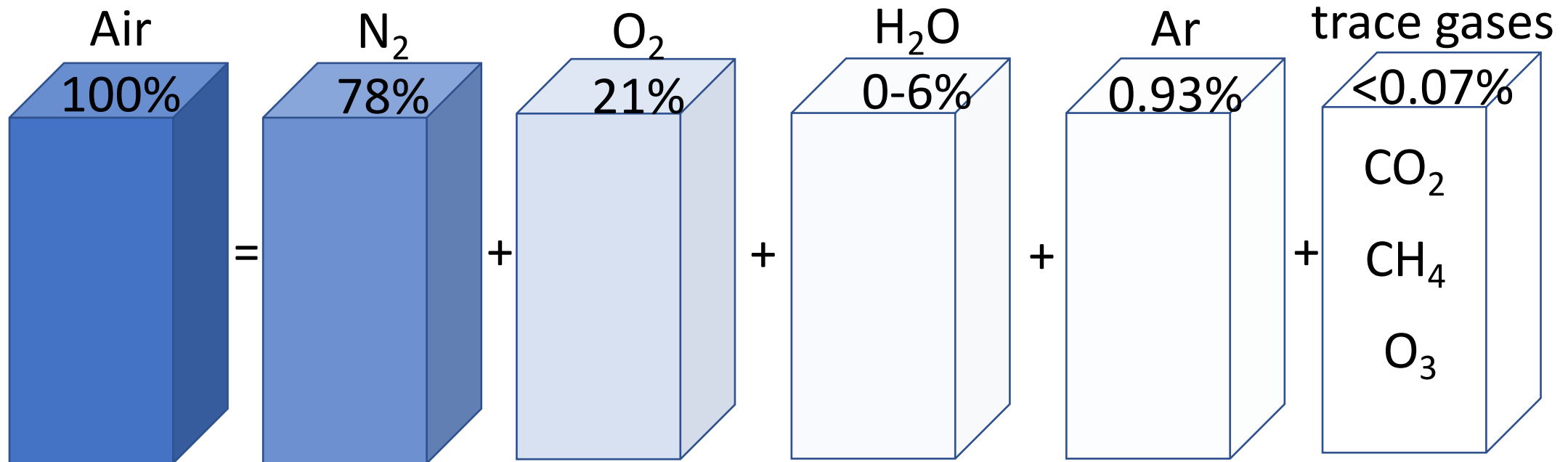


Which of these gases are inert? (or mostly inert?)

Composition of Earth's atmosphere today

Key concepts:

- What is “air”?
-
- Why don't these numbers really add up to 100%
-



Mixing ratio / number density / mass density

mixing ratio == **mole fraction** == number of moles of X per mole of air
 $\cong$ volume of gas X per volume of air

e.g. 7.9% N_2 , 430 ppb CO_2

number density == **concentration** == number of molecules of X per unit
 volume of air

e.g. Spectroscopy $\uparrow$

mass density == **mass concentration** == mass of X per unit volume of air



Important atmospheric properties

- What is a continuum fluid? *no local interactions, uniform distribution*
pressure, temperature, density are continuous as statistical averages
- What properties of our atmosphere let us treat air as a continuum fluid?



- What do we remember about temperature?



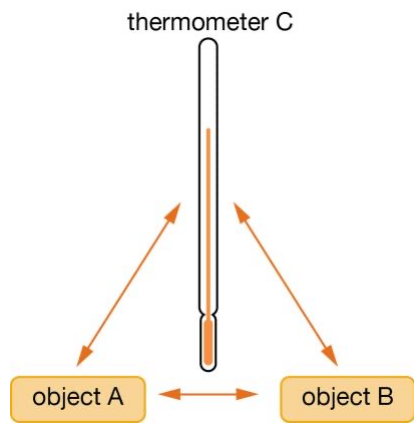
motion of molecules

- What does Local Thermodynamic Equilibrium mean?

Small gradient

7

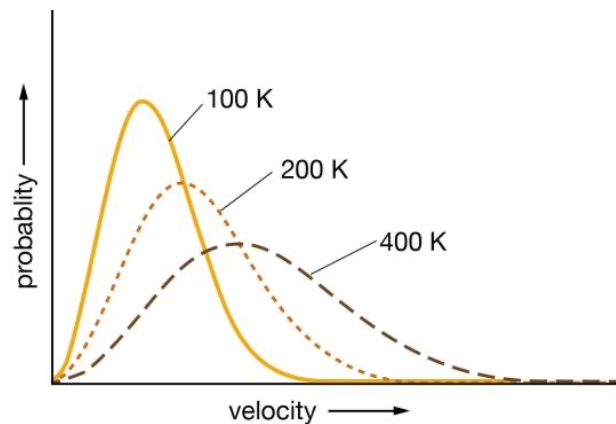
Zeroth Law of Thermodynamics



What is the 0th law?

equality

8



What is the figure on the left and what does it tell us?

velocity function

boltzmann's distribution



Key concept review

- Tell me some important properties of our Earth!

- What do you remember about “atmosphere 1”? (ie. proto-Earth)

- Talk through the evolution of Earth’s atmosphere 4.5-2.5 billion years ago.

- What big thing happened 2.5-ish billion years ago?

- Why so much N_2 vs O_2 ? Where is the Argon from?

- What are the main components of our current atmosphere? (try to remember the big percentages without going back to look)

- Why do we do a weird thing where the composition percentages don’t add to 100%

- What is mole fraction vs number density? (if I say ‘concentration’, what do I mean? If I say ‘mixing ratio’, what do I mean?)

- What is a continuum fluid and why can we treat (most of) the atmosphere as one?

- What is temperature?
